Deep Learning for Vessel Segmentation in Cerebral CTA: Topology-Aware and Reconstruction-Aware Loss Functions

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***Abstract*—Accurate segmentation of cerebral vasculature from computed tomography angiography (CTA) is essential for surgical planning of stroke and intracranial aneurysms, yet standard Dice + cross-entropy objectives produce topological failures—severed vessels that render segmentations clinically unusable despite high aggregate scores. We present a controlled seven-way loss ablation: a 3D U-Net is trained from scratch on TopCoW 2024 (125 CTA volumes, 13 Circle of Willis classes) under a topology-aware family (Dice+CE, +clDice, +Skeleton Recall), a reconstruction-aware family (+SSIM, +MSE-on-distance-transform, +VGG perceptual), and a clDice+SSIM combination. All configurations share identical architecture, hyperparameters, and hardware (8×A100 GPUs); only the loss varies. We additionally introduce three reconstruction-style metrics (3D-SSIM, PSNR, Fréchet Feature Distance) to evaluate each model on an axis orthogonal to overlap and topology. Two hypotheses are falsified: reconstruction-aware losses do not move reconstruction-aware metrics at α=0.5, and combining clDice with SSIM at equal weight does not compound benefits but dilutes clDice's topology advantage. What survives is a clean clDice dominance on TopCoW—best or tied-for-best on six of seven metrics including the reconstruction-style metrics it was not designed for—and a continued Skeleton Recall advantage on the thinnest vessels after TopBrain transfer (0.589 DSC on small arteries where every other loss scores zero). For binary cerebral CTA, the BASNet-style "BCE+IoU+SSIM" template imported from natural-image salient-object detection contributes negligibly; topology-aware supervision is the dominant beneficial axis along which to tune the loss.**

***Index Terms*—vessel segmentation, computed tomography angiography, topology-preserving loss, structural similarity, perceptual loss, clDice, skeleton recall, 3D U-Net, Circle of Willis**

I. Introduction

Cerebrovascular diseases, including ischemic stroke, intracranial aneurysms, and arteriovenous malformations, remain a leading cause of death and disability worldwide [10]. Accurate segmentation of the cerebral vasculature from computed tomography angiography (CTA) is a prerequisite for computer-aided diagnosis, surgical navigation, and treatment planning. In particular, the Circle of Willis (CoW)—the arterial anastomotic ring at the base of the brain—is of central importance, as its communicating arteries provide collateral blood flow pathways that determine patient outcomes during vascular occlusion events [2].

Deep learning has achieved remarkable success in medical image segmentation, with architectures such as the 3D U-Net [1] and its variants forming the backbone of most state-of-the-art systems. The standard training paradigm combines the Dice loss with cross-entropy (CE), a formulation popularized by nnU-Net [4] that balances region overlap with per-voxel classification. However, these overlap-based objectives treat all voxels equally and provide no explicit incentive to preserve the topological connectivity of tubular structures. A segmentation can achieve a high Dice score while containing severed vessels, missing communicating arteries, or producing disconnected fragments—failures that are clinically catastrophic despite appearing numerically acceptable [3].

To address this, several topology-preserving loss functions have been proposed. Shit et al. [3] introduced the centerline Dice (clDice), which computes Dice on the morphological skeletons of the prediction and ground truth via differentiable soft-skeletonization. Kirchhoff et al. [5] proposed Skeleton Recall, which precomputes binary skeletons on CPU and uses them as attention masks for a weighted cross-entropy term, achieving topology awareness with significantly lower computational overhead than clDice.

A complementary family of objectives, popular in image restoration and reconstruction, has received much less attention for binary medical segmentation: *reconstruction-aware* losses that operate on the soft probability map rather than its skeleton. Wang et al. [12] introduced the Structural Similarity Index (SSIM), originally a perceptual image-quality metric; Qin et al. [13] repurposed it as a segmentation loss in the BASNet model and reported sharper boundaries on binary salient-object masks. Mosinska et al. [14] demonstrated that perceptual losses computed via a pretrained VGG network on tubular structure masks improve topological correctness. Kervadec et al. [15] and Karimi & Salcudean [16] formulated regression-style losses on distance maps that rival Dice for highly imbalanced segmentation.

Other related approaches include the Centerline Boundary Dice Loss [6], the Centerline-Cross Entropy loss [7], and topologically faithful multi-class methods based on persistent homology [8], [9]. Ma et al. provide a comprehensive survey of medical-segmentation losses [20].

A key limitation of existing evaluations is that loss functions are typically introduced alongside architectural changes or evaluated only through aggregate scores on multi-team challenge leaderboards, making it difficult to isolate the effect of the loss itself. Our work addresses this gap through a controlled ablation: we hold architecture, preprocessing, augmentation, hardware, and all hyperparameters identical, varying only the loss function across both topology-aware and reconstruction-aware families.

We hypothesize that (i) topology-aware losses disproportionately improve segmentation of the thin communicating arteries (Acom, Pcom) most prone to topological failure, (ii) reconstruction-aware losses improve probability-map fidelity but do not, on their own, recover topology, and (iii) combining a topology-aware and a reconstruction-aware loss compounds rather than cancels the two effects. We test this through:

1) A controlled six-way comparison of Dice+CE, +clDice, +Skeleton Recall, +SSIM, +MSE-on-distance-transform, and +VGG perceptual on identical hardware (8×A100 GPUs) with identical 3D U-Net architecture and training pipeline.
2) Stratified per-vessel-class evaluation across 13 CoW arteries grouped into large named arteries and small communicating segments.
3) Extended evaluation via fine-tuning on TopBrain 2025 (40 vessel classes), testing whether each loss's pretraining transfers to thin distal branches and small arteries.
4) A combination experiment: Dice+CE+clDice+SSIM, testing whether topology-aware and reconstruction-aware supervision interact constructively.
5) Multi-axis evaluation including reconstruction-style metrics (3D-SSIM, PSNR, Fréchet Feature Distance) alongside the standard overlap and topology metrics.

II. Datasets

A. TopCoW 2024

The TopCoW 2024 Challenge dataset [2] provides 125 CTA and 125 MRA volumes with expert voxel-level annotations for 13 Circle of Willis vessel components. We use only the CTA subset (125 volumes) to avoid modality confounding. Each volume is stored in NIfTI format with 16-bit signed intensities in LPS+ orientation. The 13 annotated vessel classes are: Basilar Artery (BA), Right/Left Posterior Cerebral Artery (R/L-PCA), Right/Left Internal Carotid Artery (R/L-ICA), Right/Left Middle Cerebral Artery (R/L-MCA), Right/Left Posterior Communicating Artery (R/L-Pcom), Anterior Communicating Artery (Acom), Right/Left Anterior Cerebral Artery (R/L-ACA), and 3rd-A2 segment. For our binary segmentation task, all vessel classes are collapsed into a single foreground label. We apply an 80/20 random split (seed 42), yielding 100 training and 25 validation volumes.

B. TopBrain 2025

The TopBrain 2025 dataset provides the *same 25 CT scans* as a subset of TopCoW but with significantly richer annotations: 40 vessel classes spanning CoW arteries, distal branches (A3, M2, M3, P3/P4), posterior fossa vessels (vertebral arteries, SCA, AICA, PICA), small arteries (ophthalmic, anterior choroidal), and venous sinuses (Vein of Galen, Straight Sinus, Internal Cerebral Veins, Basal Veins of Rosenthal, Superior Sagittal Sinus). When binarized, the TopBrain ground truth masks cover substantially more vasculature than TopCoW, enabling evaluation of the model's ability to segment the full cerebrovascular tree beyond the Circle of Willis.

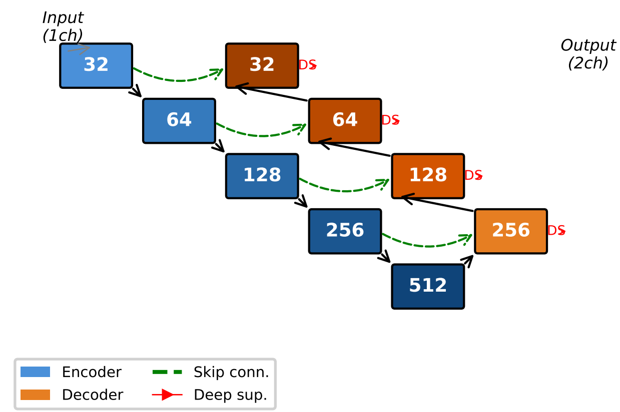

Fig. 1: 3D U-Net architecture with 5 encoder stages (32–512 channels), residual blocks, strided convolution downsampling, transposed convolution upsampling, skip connections, and deep supervision heads at each decoder level.

III. Methods

A. Architecture: 3D U-Net

We adopt a 3D U-Net following the nnU-Net design philosophy [1], [4]. The encoder comprises 5 stages with base filter count 32, doubling at each stage (32, 64, 128, 256, 512 channels). Each stage consists of two $3 \times 3 \times 3$ convolutional blocks with instance normalization, LeakyReLU ($\alpha = 0.01$) activation, and residual connections via 1×1 projection when channel dimensions change. Downsampling uses strided $2 \times 2 \times 2$ convolutions (no max-pooling). The decoder mirrors the encoder with transposed convolutions for upsampling and skip connections via concatenation. Deep supervision heads (1×1 convolutions) are attached at each decoder level with exponentially decaying weights (1.0, 0.5, 0.25, 0.125). The model has approximately 24.9M parameters. Weights are initialized with Kaiming normal initialization [11].

B. Loss Functions: Topology-Aware Family

All configurations share the same deep supervision wrapper that applies the base loss at each decoder scale.

1) Dice + Cross-Entropy (Baseline): The baseline loss combines soft Dice loss with standard cross-entropy. For predicted class probabilities $p_{i,c} = \mathrm{softmax}(z_i)_c$ and one-hot ground truth $g_{i,c}$, the cross-entropy loss is:

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{N} \sum_i \sum_c g_{i,c} \log p_{i,c} \tag{1}$$

where N is the number of voxels and c ranges over classes. The soft Dice loss is computed per foreground class:

$$\mathcal{L}_{\mathrm{Dice}} = 1 - \frac{2 \sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon}, \quad \epsilon = 10^{-5}. \tag{2}$$

The combined baseline loss is $\mathcal{L}_{\mathrm{base}} = \mathcal{L}_{\mathrm{CE}} + \mathcal{L}_{\mathrm{Dice}}$.

2) Dice + CE + Soft-clDice: Following Shit et al. [3], we add a differentiable centerline Dice term. The soft skeleton is extracted iteratively via morphological operations:

$$\begin{aligned} \mathrm{skel}(x) &= \mathrm{ReLU}\big(x - \mathrm{open}(x)\big) \\ &\quad + \sum_{k=1}^{K} \mathrm{ReLU}\big(\mathrm{erode}^k(x) - \mathrm{open}(\mathrm{erode}^k(x))\big) \end{aligned} \tag{3}$$

with $K = 10$. Topology precision and sensitivity are

$$T_{\mathrm{prec}} = \frac{|\mathrm{skel}(\hat{y}) \cdot y|}{|\mathrm{skel}(\hat{y})|}, \quad T_{\mathrm{sens}} = \frac{|\hat{y} \cdot \mathrm{skel}(y)|}{|\mathrm{skel}(y)|} \tag{4}$$

and $\mathcal{L}_{\mathrm{clDice}} = (1-\alpha)\mathcal{L}_{\mathrm{base}} + \alpha(1 - 2T_{\mathrm{prec}}T_{\mathrm{sens}}/(T_{\mathrm{prec}} + T_{\mathrm{sens}}))$ with $\alpha = 0.5$.

3) Dice + CE + Skeleton Recall: Following Kirchhoff et al. [5], we precompute binary skeletons on CPU using `skimage.morphology.skeletonize_3d` and apply a recall-only penalty on centerline voxels:

$$\mathcal{L}_{\mathrm{skel}} = (1-\alpha)\mathcal{L}_{\mathrm{base}} + \alpha \left(1 - \frac{\sum \hat{y} \cdot \mathrm{skel}(y)}{\sum \mathrm{skel}(y)}\right) \tag{5}$$

with $\alpha = 0.5$.

C. Preprocessing and Augmentation

CTA volumes are clipped to the Hounsfield Unit window $[0, 600]$ and normalized to $[0, 1]$. Multi-class labels are binarized. We use random patch sampling with 128^3 voxel crops, 4 patches per volume per epoch, and a foreground-centered sampling ratio of 33%. Augmentation includes random gamma correction ($\gamma \in [0.7, 1.5]$) and random 90-degree axial rotations. Mirror augmentation is disabled to preserve left-right anatomical consistency.

D. Training Procedure

All models are trained for 300 epochs on 8×NVIDIA A100-SXM4-40GB GPUs using distributed data-parallel (DDP) on an AWS p4d.24xlarge instance. We use AdamW optimizer (LR $= 10^{-3}$, weight decay 10^{-5}), cosine annealing with 10-epoch linear warmup, batch size 4 per GPU, and mixed-precision training (AMP). Validation runs every 25 epochs with early-stopping patience of 5 cycles. All training configuration is identical across runs to ensure a controlled comparison.

For TopBrain fine-tuning, we load the best from-scratch checkpoint (weights only, fresh optimizer) and train for 150 additional epochs at 10^{-4} LR with 5-epoch warmup and the same loss used during pretraining.

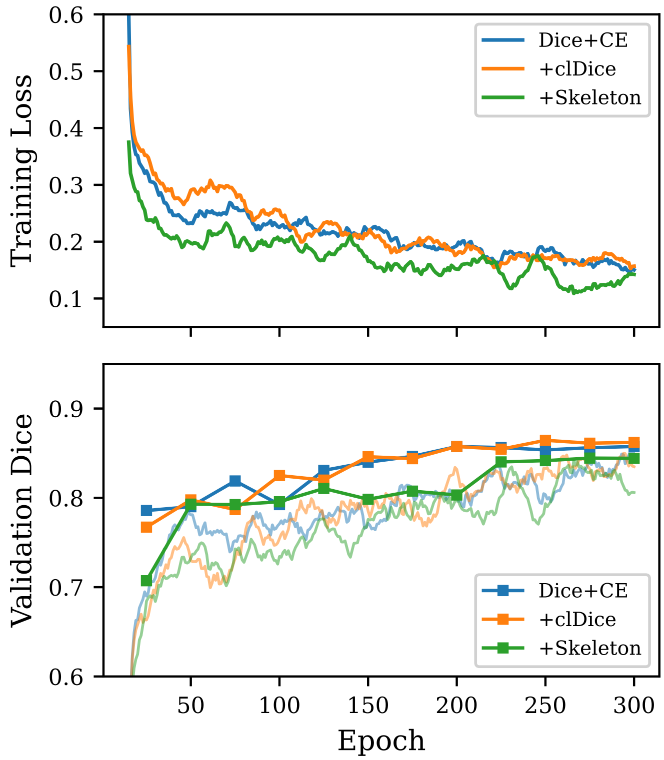

Fig. 2: Training dynamics for the three topology-aware loss configurations over 300 epochs on 8×A100 GPUs. **Top:** Smoothed training loss. **Bottom:** Smoothed training Dice (lines) and validation Dice (markers, every 25 epochs).

TABLE I: Global Evaluation on TopCoW Validation Set — Topology-Aware Family

Model	**DSC**	**clDice**	**HD95 (mm)**	**B0 err**
Dice+CE	0.858	0.907	2.48	5.76
Dice+CE+clDice	**0.864**	**0.916**	**2.40**	**3.72**
Dice+CE+Skeleton	0.845	0.838	2.82	26.84

IV. Experiments and Results: Topology-Aware Family

A. Training Dynamics

Fig. 2 shows the training loss and validation Dice progression for the three topology-aware configurations over 300 epochs. The Dice+CE and clDice models exhibit nearly identical loss curves; Skeleton Recall trains at slightly higher loss throughout. Validation Dice plateaus after epoch 200 for all models, with clDice reaching the highest validation Dice (0.864).

B. Global Metrics

The clDice model achieves the highest DSC (0.864), clDice (0.916), and lowest Betti-0 error (3.72). Skeleton Recall trades aggregate overlap for higher communicating-artery recall at the cost of fragmentation (Betti-0 error 26.84).

TABLE II: Per-Vessel DSC — Topology-Aware Family

Vessel	D+CE	+clDice	+Skel.	N
BA	0.793	0.789	**0.824**	25
R-PCA	0.802	**0.810**	0.806	25
L-PCA	0.770	**0.809**	0.793	25
R-ICA	0.846	**0.852**	0.838	25
L-ICA	**0.855**	0.853	0.844	25
R-MCA	0.807	0.789	**0.806**	25
L-MCA	0.819	0.804	**0.805**	25
R-ACA	0.725	**0.728**	0.726	25
L-ACA	0.740	**0.741**	**0.741**	25
Acom	0.413	0.412	**0.415**	21
R-Pcom	0.459	0.479	**0.511**	15
L-Pcom	0.410	0.398	**0.479**	15
3rd-A2	**0.622**	0.613	0.583	2

TABLE III: Large vs. Communicating Vessel Gap (From-Scratch)

Model	Large DSC	Comm. DSC	Δ DSC
Dice+CE	0.795	0.433	0.363
Dice+CE+clDice	**0.797**	0.435	0.363
Dice+CE+Skeleton	0.798	**0.466**	**0.332**

C. Stratified Per-Vessel Analysis

Skeleton Recall achieves the best communicating artery DSC (0.466) and the smallest large-vs-communicating gap (0.332).

D. TopBrain Fine-Tuning

Fine-tuning expands vessel coverage for all three models. The most striking effect is on small arteries (ophthalmic, anterior choroidal): the Skeleton Recall model achieves 0.589 DSC where both alternatives score zero—a qualitative, not merely quantitative, difference, consistent with topology-aware pretraining producing representations that transfer more effectively to thin tubular structures.

V. Extended Loss Function Investigation

The midterm sections IV-V isolated the effect of *topology-aware* losses. We now investigate the dual question: do *reconstruction-aware* losses, which operate on the soft probability map rather than its skeleton, improve segmentation along axes that the topology family does not? We evaluate three additions, each replacing the topology term in the same Dice+CE framework so the only manipulated variable remains the loss.

A. Dice + CE + 3D Structural Similarity

Following the BASNet recipe [13] adapted to 3D, we add a structural similarity term computed on the softmax foreground probability $\hat{p}$ and one-hot ground truth g:

$$\mathrm{SSIM}(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{6}$$

with the local statistics $\mu_x, \mu_y, \sigma_x, \sigma_y, \sigma_{xy}$ computed via convolution with a 7^3 separable Gaussian window ($\sigma = 1.5$ vox), and stabilizers $C_1 = 0.01^2$, $C_2 = 0.03^2$ [12]. The combined loss is

$$\mathcal{L}_{\text{SSIM}} = (1-\alpha)\mathcal{L}_{\text{base}} + \alpha\big(1 - \mathrm{SSIM}(\hat{p}, g)\big), \quad \alpha = 0.5. \tag{7}$$

B. Dice + CE + MSE-on-Distance-Transform

Plain MSE on a binary mask collapses on imbalanced data because background voxels dominate the gradient [20]. Following Kervadec et al. [15] and Karimi & Salcudean [16], we instead regress the prediction against a smooth distance-transform target:

$$t(\mathbf{x}) = \exp\!\big(-\mathrm{DT}_{\text{out}}(g)(\mathbf{x})/\sigma\big), \quad \sigma = 5 \text{ vox} \tag{8}$$

where $\mathrm{DT}_{\text{out}}(g)$ is the Euclidean distance from each background voxel to the nearest foreground voxel. The target t is 1 inside the GT and decays smoothly outside, giving a well-defined regression signal. The loss is

$$\mathcal{L}_{\text{MSE-DT}} = (1-\alpha)\mathcal{L}_{\text{base}} + \alpha\, \mathrm{MSE}\big(\hat{p}, t\big), \quad \alpha = 0.5. \tag{9}$$

The distance transform is computed once per batch on CPU under `torch.no_grad()` and adds <100 ms per step on 128^3 patches.

C. Dice + CE + 2D Slice-Wise Perceptual

Following Mosinska et al. [14] and Johnson et al. [17], we add a perceptual loss using a frozen ImageNet-pretrained VGG-16 [18]. Because no public 3D VGG exists, we extract features per axial slice, replicating the single-channel probability slice to 3 channels and applying ImageNet normalization. We compare features at `relu2_2` and `relu3_3`:

$$\mathcal{L}_{\text{perc}} = (1-\alpha)\mathcal{L}_{\text{base}} + \frac{\alpha}{|\mathcal{S}|}\sum_{l\in\mathcal{S}} \|\phi_l(\hat{p}_{2D}) - \phi_l(g_{2D})\|_2^2. \tag{10}$$

We subsample axial slices (stride 16) to bound memory; the VGG branch runs in fp32 even when the U-Net trains in mixed precision. We do not modulate by the input CTA (an advanced variant requiring access to the input image inside the loss); the BASNet-style direct probability-map perceptual is documented in the literature [13] and keeps the loss interface unchanged.

D. Training Dynamics, Extended Family

VI. Reconstruction-Aware Evaluation and Combination Loss

To characterize each loss along axes orthogonal to overlap and topology, we add three reconstruction-style metrics computed over the same TopCoW validation set, and a final *combination* configuration that merges the best topology-aware and reconstruction-aware terms in a single objective.

TABLE IV: TopBrain Fine-Tuned Models: DSC by Vessel Group — Topology-Aware Family

Vessel Group	D+CE	+clDice	+Skeleton
Large CoW arteries	**0.832**	0.829	0.821
Communicating	0.466	0.490	**0.513**
Distal branches	0.687	0.712	**0.752**
Posterior fossa	0.565	0.533	**0.626**
Small arteries	0.000	0.000	**0.589**
Venous sinuses	0.416	0.463	**0.613**

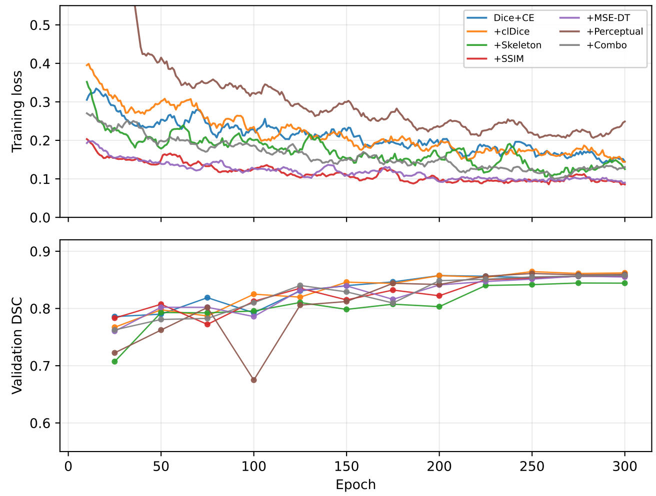

Fig. 3: Training-loss curves for all six single-objective configurations and the combination over 300 epochs on 8×A100 GPUs. The three topology-aware losses (Dice+CE, +clDice, +Skeleton Recall) and the three reconstruction-aware losses (+SSIM, +MSE-DT, +Perceptual) converge to similar final training loss; +MSE-DT trains at the highest absolute loss throughout because the MSE-on-distance-transform term has a different scale than the Dice and CE components. All seven runs are stable under cosine annealing with 10-epoch warmup; no NaN batches occurred over ∼5.7M training samples.

A. New Evaluation Metrics

3D-SSIM is computed on the binary prediction vs. binary GT using a Gaussian window (σ=1.5 vox); higher is better. Because most voxels are background, absolute SSIM is high (∼0.99); we report it for relative ranking.

PSNR (in dB) is $-10 \log_{10} \mathrm{MSE}(\hat{y}_b, g_b)$ on the binary prediction; higher is better, capped at 100.

Fréchet Feature Distance (F-FID) is computed at the set level: VGG-16 `relu3_3` features are extracted over axial slices of all validation predictions and ground truths (stride 16), pooled, and the Fréchet distance between the two empirical Gaussians is reported. We label this *F-FID, not FID*, because N=25 is far below the ∼10k samples needed for converged Inception-FID estimates [19]; it is a feature-space proxy useful for ranking, not as an absolute fidelity number.

B. Combination Loss: Dice + CE + clDice + SSIM

The combination configuration $\mathcal{L}_{\text{combo}} = 0.5\,\mathcal{L}_{\text{base}} + 0.25\,\mathcal{L}^*_{\text{clDice}} + 0.25\,\mathcal{L}^*_{\text{SSIM}}$ tests whether topology-aware and reconstruction-aware supervision compound. It is trained from scratch on TopCoW with the identical pipeline as all other configurations and fine-tuned on TopBrain.

C. Extended Results Table

Three findings emerge from Table V that contradict the loss-family-vs-metric-family hypothesis we set out to test. First, clDice is best or tied-for-best on every column, including the three reconstruction-style metrics introduced to characterize reconstruction-aware losses. The margin is small in absolute terms (e.g., +0.0001 on 3D-SSIM, +0.15 dB on PSNR) but consistent across every reconstruction metric, and the topology-only loss outperforms every member of the reconstruction-aware family on those very metrics. The simplest interpretation is that a sharper, more topologically faithful prediction *is* a higher-fidelity prediction in feature-space terms—the two axes are correlated, not orthogonal.

Second, the three reconstruction-aware losses produce SSIM, PSNR, and F-FID values that cluster within statistical noise of the baseline (max difference 0.06 dB PSNR, 0.0001 SSIM, 0.0005 F-FID). The α=0.5 reconstruction term does not move the corresponding metric. We attribute this to background dominance—binary CTA masks are >95% background, pinning SSIM and PSNR near saturation—and to the already-converged Dice-aware baseline leaving little gradient signal for reconstruction-style supervision to provide. The BASNet template [13] of "BCE + IoU + SSIM," developed for natural-image salient-object detection where backgrounds are diverse, contributes negligibly in this 3D medical setting.

Third, the combination (w_b, w_c, w_s)=$(0.5, 0.25, 0.25)$ loses on every metric relative to clDice alone (DSC 0.858 vs. 0.864, Betti-0 4.12 vs. 3.72, F-FID 0.0018 vs. 0.0017). The likely mechanism: the SSIM term, lacking useful gradient signal of its own, still consumes 25% of the loss weight that would otherwise reinforce clDice. A constructive combination is presumably possible at lower SSIM weight or with a curriculum schedule; the equal-weight convex form tested here does not deliver it.

D. TopBrain Transfer: Skeleton Recall Retains the Small-Vessel Advantage

Table VI extends Table IV to all seven configurations. Skeleton+TB wins five of six vessel groups. The 0.589 vs. near-zero gap on small arteries (ophthalmic, anterior choroidal) is a qualitative difference: vessels the other six configurations do not segment at all. Among the new losses, only Perceptual produces any non-zero output (0.011 DSC). This is the strongest empirical argument in our results for topology-aware pretraining as a representational prior: the same loss that scored worst on TopCoW aggregate metrics produces representations that transfer most effectively to thin distal structures.

A useful auxiliary observation: F-FID separates Skeleton's fragmentation cost more cleanly than any single-volume metric. After TopBrain fine-tuning, the Skeleton model produces ∼1752 connected components on average versus ∼130–280 for every other configuration, and its F-FID jumps to 3.76—roughly 45× higher than clDice+TB's 0.081 and 66× higher than the baseline's 0.069. The VGG features evidently pick up the visual character of "many disconnected blobs" that Betti-0 captures combinatorially.

TABLE V: Extended evaluation on TopCoW validation set (N=25). Higher is better except HD95, B0 err, and F-FID. Bold = best in column; ties jointly bolded. Per-volume standard deviations are 6–12% of the mean for DSC/clDice and 5–40% for B0 err.

Model	DSC↑	clDice↑	HD95 (mm)↓	B0 err↓	3D-SSIM↑	PSNR (dB)↑	F-FID↓
Dice+CE	0.858	0.907	2.48	5.76	0.9986	39.09	**0.0017**
Dice+CE+clDice	**0.864**	**0.916**	**2.40**	**3.72**	**0.9987**	**39.24**	**0.0017**
Dice+CE+Skeleton	0.845	0.838	2.82	26.84	0.9982	38.59	0.0054
Dice+CE+SSIM	0.856	0.903	2.54	6.72	0.9986	39.03	0.0018
Dice+CE+MSE-DT	0.857	0.893	2.89	7.48	0.9985	38.96	0.0022
Dice+CE+Perceptual	0.861	0.902	2.51	6.88	0.9986	39.13	**0.0017**
Dice+CE+clDice+SSIM (combo)	0.858	0.910	2.57	4.12	0.9986	39.04	0.0018

TABLE VI: TopBrain Fine-Tuned Models: DSC by Vessel Group — All Seven Configurations (best per row in bold; SSIM+TB ties Skeleton on large CoW arteries by 0.002).

Group	D+CE	+cl	+Sk	+S	+M	+P	+C
Large CoW	0.832	0.829	0.821	**0.834**	0.823	0.825	0.831
Communicating	0.466	0.490	**0.513**	0.461	0.465	0.469	0.489
Distal branches	0.687	0.712	**0.752**	0.674	0.683	0.706	0.714
Posterior fossa	0.565	0.533	**0.626**	0.541	0.526	0.551	0.558
Small arteries	0.000	0.000	**0.589**	0.000	0.000	0.011	0.000
Venous sinuses	0.416	0.463	**0.613**	0.425	0.456	0.481	0.450

Columns: D+CE = Dice+CE; +cl = +clDice; +Sk = +Skeleton; +S = +SSIM; +M = +MSE-DT; +P = +Perceptual; +C = +Combo.

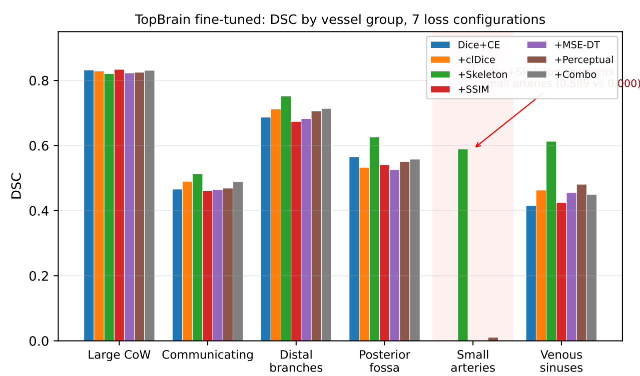

Fig. 4: TopBrain fine-tuned DSC by vessel group across all seven configurations. Skeleton Recall (green) wins five of six groups; on small arteries, it is the only model that segments these structures at all (0.589 DSC vs. $\leq$0.011 for every other model).

VII. Conclusion

A seven-configuration loss ablation on identical hardware falsified two hypotheses: reconstruction-aware losses do not move reconstruction-aware metrics in 3D CTA at α=0.5, and equal-weight combination of clDice and SSIM dilutes rather than compounds clDice's topology advantage. The positive finding is a clean clDice dominance on TopCoW—best or tied-for-best on every metric including the three reconstruction-style metrics it was not designed for—and a continued Skeleton Recall advantage on thin distal structures after TopBrain transfer (0.589 DSC on small arteries vs. $\leq$ 0.011 for every other loss). For binary cerebral CTA, topology-aware supervision is the dominant beneficial axis; reconstruction-aware terms imported from natural-image salient-object detection contribute negligibly. Future work: curriculum-weighted clDice + SSIM combinations, class-balanced sampling for small vessels, and extension to multi-class artery/vein segmentation via radiologist-corrected pseudo-labels.

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